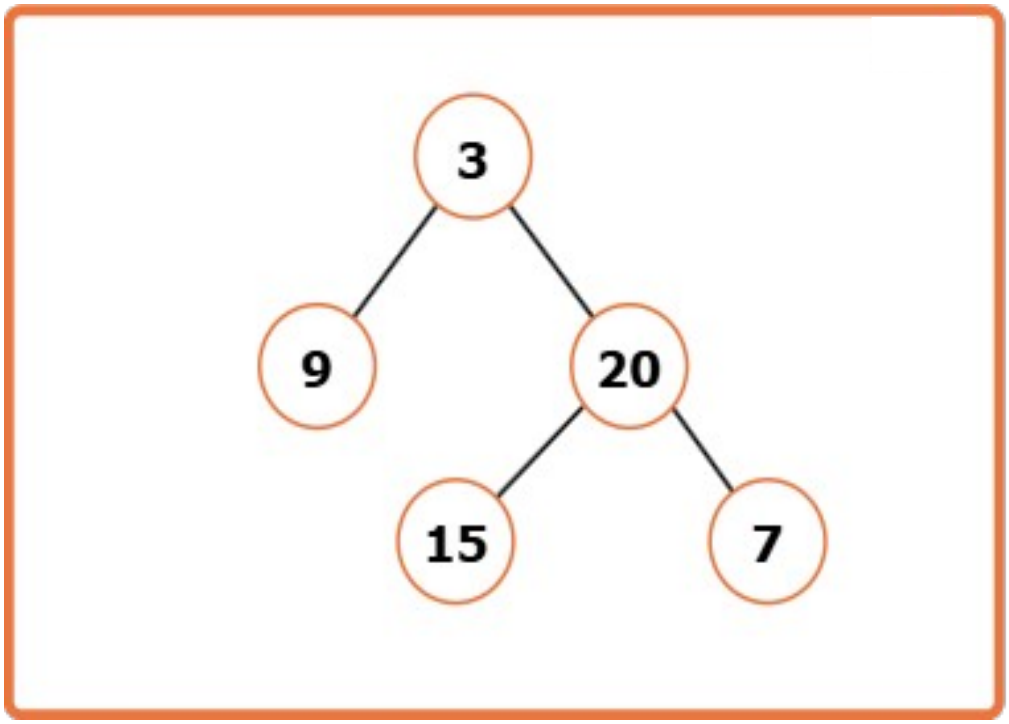
Construct A Binary Tree from Inorder and Preorder Traversal

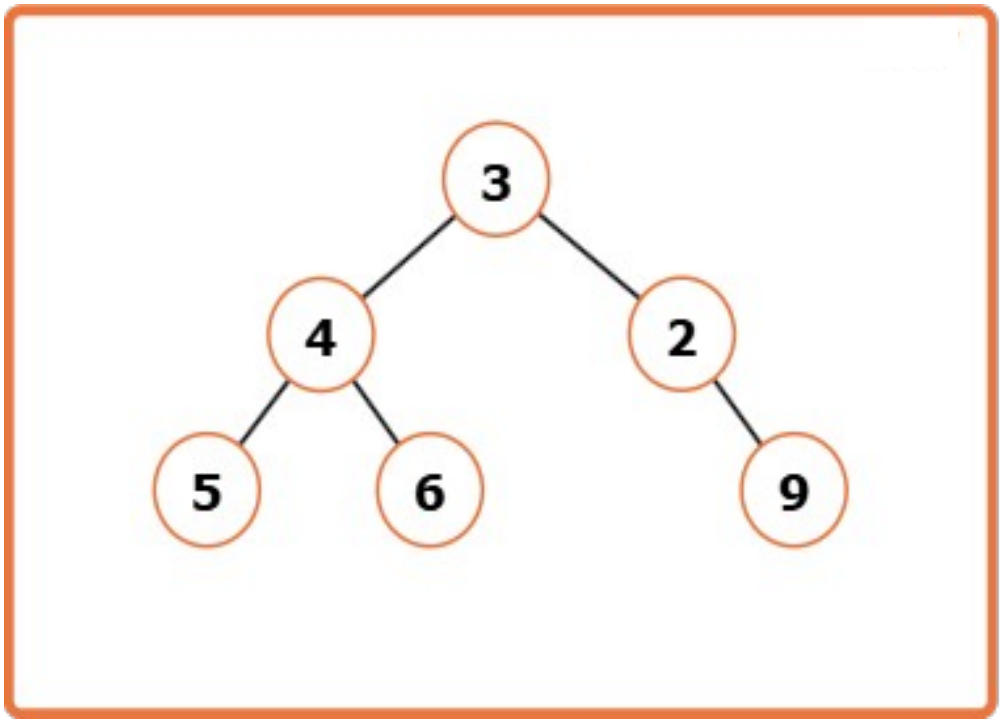
Problem Statement: Given the Preorder and Inorder traversal of a Binary Tree, construct the Unique Binary Tree represented by them..

Examples

```
Input : preorder = [3, 9, 20, 15, 7] , inorder = [9, 3, 15, 20, 7]
Output : [3, 9, 20, null, null, 15, 7]
Explanation : The output tree is shown below.
```




```
Input : preorder = [3, 4, 5, 6, 2, 9] , inorder = [5, 4, 6, 3, 2, 9]
Output : [3, 4, 2, 5, 6, null, 9]
Explanation : The output tree is shown below.
```



Approach
Algorithm

To construct a binary tree from preorder and inorder traversals, we utilize two key properties:

- **Preorder traversal** gives us the root of the tree/subtree first.
- **Inorder traversal** helps determine the left and right subtrees by locating the root's index.

With these insights, we can recursively build the tree. Using indices instead of slicing arrays avoids extra space, and a hashmap helps us find root positions in `O(1)` time.

- Store each value's index from the inorder traversal in a hashmap for quick lookup.
- Create a recursive function `buildTree` that takes preorder, inorder, and their current index boundaries.
- Base case: If the boundaries are invalid (preStart > preEnd or inStart > inEnd), return NULL.
- Identify the root as `preorder[preStart]`, and find its index in the inorder traversal using the hashmap.
- Calculate the size of the left subtree: `leftSubtreeSize = rootIndex - inStart`.
- Recursively build the left subtree using updated indices based on the leftSubtreeSize.
- Recursively build the right subtree using the remaining indices.
- Return the root node with left and right children connected.

Code

C++

```
#include <bits/stdc++.h>
using namespace std;

// TreeNode structure
struct TreeNode {
    int val;
    TreeNode *left;
    TreeNode *right;

    // Constructor to initialize a node
    TreeNode(int x) : val(x), left(nullptr), right(nullptr) {}
};

class Solution {
public:
    // Function to build the binary tree from preorder and inorder
    TreeNode* buildTree(vector<int>& preorder, vector<int>& inorder) {
        // Map to store the index of each value in inorder
        map<int, int> inMap;

        // Fill the map with inorder values and their indices
        for (int i = 0; i < inorder.size(); i++) {
            inMap[inorder[i]] = i;
        }

        // Call the recursive helper function
        return build(preorder, 0, preorder.size() - 1, inorder, 0,
inorder.size() - 1, inMap);
    }

private:
```

```
    // Recursive function to build tree using preorder and inorder segments
    TreeNode* build(vector<int>& preorder, int preStart, int preEnd,
                    vector<int>& inorder, int inStart, int inEnd, map<int, int>&
inMap) {

        // Base condition
        if (preStart > preEnd || inStart > inEnd) return nullptr;

        // The first element in preorder is root
        TreeNode* root = new TreeNode(preorder[preStart]);

        // Find the root index in inorder
        int inRoot = inMap[root->val];

        // Number of elements in left subtree
        int numsLeft = inRoot - inStart;

        // Recursively build left and right subtrees
        root->left = build(preorder, preStart + 1, preStart + numsLeft,
                           inorder, inStart, inRoot - 1, inMap);
        root->right = build(preorder, preStart + numsLeft + 1, preEnd,
                            inorder, inRoot + 1, inEnd, inMap);

        return root;
    }
};

// Inorder traversal to print tree
void printInorder(TreeNode* root) {
    if (!root) return;
    printInorder(root->left);
    cout << root->val << " ";
    printInorder(root->right);
}

int main() {
    vector<int> inorder = {9, 3, 15, 20, 7};
    vector<int> preorder = {3, 9, 20, 15, 7};

    Solution sol;
    TreeNode* root = sol.buildTree(preorder, inorder);

    cout << "Inorder of Unique Binary Tree Created:\n";
    printInorder(root);
    cout << endl;

    return 0;
}
```

JAVA

```
// TreeNode structure
class TreeNode {
    int val;
    TreeNode left, right;

    // Constructor
    TreeNode(int x) {
        val = x;
        left = right = null;
    }
}
```

```
// Solution class
class Solution {
    // Public method to start the building process
    public TreeNode buildTree(int[] preorder, int[] inorder) {
        // Map to store value -> index from inorder
        Map<Integer, Integer> inMap = new HashMap<>();
        for (int i = 0; i < inorder.length; i++) {
            inMap.put(inorder[i], i);
        }

        // Start the recursive construction
        return build(preorder, 0, preorder.length - 1, inorder, 0,
inorder.length - 1, inMap);
    }

    // Helper method
    private TreeNode build(int[] preorder, int preStart, int preEnd,
                           int[] inorder, int inStart, int inEnd, Map<Integer,
Integer> inMap) {
        // Base condition
        if (preStart > preEnd || inStart > inEnd) return null;

        // First element of preorder is the root
        TreeNode root = new TreeNode(preorder[preStart]);

        // Get inorder index of root
        int inRoot = inMap.get(root.val);
        int numsLeft = inRoot - inStart;

        // Build left and right subtrees
        root.left = build(preorder, preStart + 1, preStart + numsLeft,
                          inorder, inStart, inRoot - 1, inMap);
        root.right = build(preorder, preStart + numsLeft + 1, preEnd,
                           inorder, inRoot + 1, inEnd, inMap);

        return root;
    }
}

// Main class
public class Main {
    // Method to print inorder traversal
    public static void printInorder(TreeNode root) {
        if (root == null) return;
        printInorder(root.left);
        System.out.print(root.val + " ");
        printInorder(root.right);
    }

    public static void main(String[] args) {
        int[] inorder = {9, 3, 15, 20, 7};
        int[] preorder = {3, 9, 20, 15, 7};

        Solution sol = new Solution();
        TreeNode root = sol.buildTree(preorder, inorder);

        System.out.println("Inorder of Unique Binary Tree Created:");
        printInorder(root);
    }
}
```

Time Complexity: O(N), as ach node is visited once. **Space Complexity:** O(N), for the hashmap and recursion stack (worst case when tree is skewed).